Lecture 09 - Network Layer-IPv4 Functions

Summer 2025 (A)

7.I. How does a device receiving an ICMP packet know that it is a PING Echo request?

II. When running traceroute, why do you sometimes see * * * instead of a router's reply?

III. State one key difference between a datagram network and a virtual circuit network.

I. How does a device know an ICMP packet is a PING Echo request?

A device identifies a **PING Echo request** by examining the **ICMP header**.

- **ICMP Type field = 8** → Echo Request
- **ICMP Code field = 0**

When a device receives an ICMP packet with **Type 8**, it knows the packet is a **PING Echo request** and replies with:

- **ICMP Type 0** → Echo Reply

II. Why does traceroute sometimes show * * * instead of a router's reply?

* * * appears when **no ICMP response is received within the timeout period**.

Common reasons:

- The router is configured to **block or ignore ICMP packets**
- **Firewall or security policies** drop ICMP Time Exceeded messages

- **Network congestion** causes packet loss or delayed replies
- The router is **overloaded** and does not respond

Each * represents a **missed reply for one probe**.

III. One key difference between a datagram network and a virtual circuit network

Feature	Datagram Network	Virtual Circuit Network
Setup Phase	None (Connectionless)	Required (Connection-oriented)
Packet Path	Can vary for each packet	Fixed for the entire session
Addressing	Full source/destination address in each packet	Short Virtual Circuit Identifier (VCI)
State Info	Routers do not keep state of connections	Routers maintain state for each VC

Example (for understanding):

- Internet (IP) → Datagram network
- ATM / Frame Relay → Virtual circuit network

Spring 2025 (A)

6.A university lab computer can successfully ping another laptop in the same building, but it fails when trying to ping a server on the internet.

I. **Give two possible reasons** why the computer cannot ping the internet server.

II. Can a network admin use ICMP tools to determine whether the problem is inside or outside the campus network? Explain which ICMP tools are used to determine the external and internal issues.

Scenario:

- **Lab computer** can ping **local laptop** → local network is fine
- Cannot ping **internet server** → problem with external connectivity

I. Two possible reasons the computer cannot ping the internet server

1. **Incorrect or missing default gateway**
    - If the PC doesn't know the router to reach outside networks, packets to the internet cannot be delivered.
2. **Firewall or network restrictions**
    - Either on the **PC**, the **campus firewall**, or the **ISP** blocking ICMP packets (ping).

Other possible reasons:

- DNS resolution issue (if ping by hostname)
- ISP or router outage

II. Using ICMP tools to locate the problem

Yes, a network admin can use **ICMP-based diagnostic tools**:

1. **Ping**
    - Used to check **reachability** of devices.
    - Ping a **local router or default gateway** → tests **internal network**.
    - Ping an **external IP (e.g., 8.8.8.8)** → tests **external connectivity**.
2. **Traceroute**
    - Shows the **path packets take** to reach a destination.
    - Helps determine **where the failure occurs**:
        - If failure occurs **within first few hops** → problem **inside campus network**
        - If failure occurs **after leaving campus network** → problem **outside**

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7. In the IPv4 header, the Total Length field is 16 bits, while the Fragment Offset field is only 13 bits.
I. **How** does the 13-bit Fragment Offset represent the position of a fragment accurately without losing information?
II. **What** is the purpose of the Identification field in fragmentation and reassembly?

I. How does the 13-bit Fragment Offset represent the position accurately?

- **Fragment Offset field = 13 bits** → can represent values **0 to 8191**
- **Unit of measurement**: **8 bytes (64 bits)**, not 1 byte
- Maximum offset in **bytes** = 8191 × 8 = **65528 bytes**
- This allows the **position of each fragment** in the original packet to be known, even though the total length field can go up to **65535 bytes**.

Key idea: Using **8-byte units**, 13 bits are enough to represent the position of fragments in a large packet.

II. Purpose of the Identification field

- Each original IPv4 packet has a **unique Identification number**
- All fragments of that packet carry the **same Identification value**
- **Purpose:**
    - Allows the receiver to **group fragments** belonging to the same original packet
    - Enables **correct reassembly** of fragmented packets

Fall 2024 (A)

9.A customer support agent at an ISP receives a call from a client who is unable to access a critical business application hosted externally. The client's network appears to be fine when accessing other internet services. **Identify** which diagnostic tool the agent should use to pinpoint where the connectivity issue to the business application is occurring. **Explain** what specific information the tool provides that will help identify the exact location of the problem.

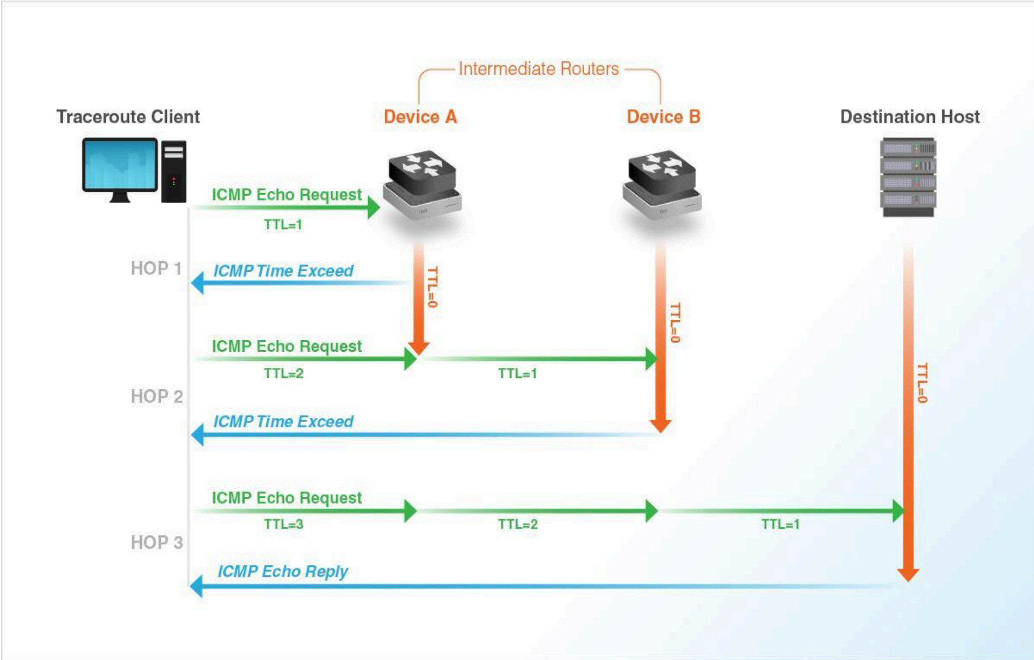
In this scenario, the most effective diagnostic tool the customer support agent should use is **Traceroute**.

While a tool like **Ping** can confirm that the application is unreachable, it cannot explain *where* the connection is breaking. Traceroute is specifically designed to map the path of a packet and reveal the exact location of a failure.

How Traceroute Works:

Traceroute sends a series of packets toward the destination with an increasing **Time-to-Live (TTL)** value.

- **The TTL Mechanism:** Every time a packet hits a router (a "hop"), the TTL value decreases by one. When it hits zero, the router discards the packet and sends back an "ICMP Time Exceeded" message.
- **The Discovery Process:** By starting with a TTL of 1, then 2, then 3, the tool forces every router along the path to identify itself in sequence until the packet finally reaches the external business application.



Information Provided by the Tool

Traceroute provides three critical pieces of data for each hop that help pinpoint the problem:

1. **Hop Number and IP/Hostname:** It lists every router the data passes through. This allows the agent to see if the connection is failing inside the **client's local network**, the **ISP's internal network**, or at an **upstream provider** (the "middle" of the internet).
2. **Round-Trip Time (RTT):** The tool typically sends three packets per hop and displays the response time in milliseconds (ms).
   - **High Latency:** If latency jumps from 20ms to 300ms at a specific hop, that router is congested or misconfigured.
   - **Packet Loss:** If a hop returns asterisks (* * *), it indicates the packet was dropped or the router is not responding.
3. **The Point of Failure:** If the traceroute suddenly stops and cannot reach the final destination, the **last successful hop** is the key.
   - If it stops at a router belonging to the business application's host, the issue is with the application provider.
   - If it stops within the ISP's infrastructure, the ISP needs to fix a routing issue.

Spring 2024 (A)

9.Mention **what kind** of information you get to see when you do a traceroute from your home PC to a YAHOO server. **How does** traceroute help us to troubleshoot?

Traceroute Information:

- List of all **routers (hops)** between my PC and the Yahoo server
- **IP addresses** of each hop

- **Round-Trip Time (RTT)** to each hop (usually 3 measurements)
- Sometimes **hostnames** of routers (if reverse DNS is available)

How traceroute helps troubleshoot:

- **Identify delays**: hops with high RTT indicate congestion or slow links
- **Detect packet loss**: hops showing `* * *` may be down or blocking traffic
- **Understand network path**: shows the route packets take to reach the server
- **Locate ISP or routing issues**: helps determine where connectivity problems occur